

Paragliding Intelligence Platform

Volume 18 - Data Analytics & Growth Intelligence v2 Technical

SkyThermal - Product Analytics, Safety Analytics, Forecast Accuracy, Growth Intelligence and Decision Intelligence Metrics

Document Field	Value
Document ID	ST-V18-DATA-GROWTH-v2
Status	Technical design baseline
Primary readers	Product, data engineering, growth, ML, operations, investor reporting, safety committee
Depends on	Volumes 01-17: PRD, SAD, DDD, API, Weather, AI/ML, UX, Infra, Security, QA, Business, Legal, Playbooks, Sensors, Rescue, Portal, Frontend
Core goal	Turn product usage, weather accuracy, safety outcomes and subscription behavior into measurable system intelligence

1. Executive Summary

This volume defines the analytics and growth intelligence architecture for SkyThermal, a paragliding-first meteo and decision-support platform. The analytics system must measure more than clicks. It must quantify whether forecasts are useful, whether flyability decisions avoid false-green conditions, whether alerts reach pilots at the correct time, whether launch-site playbooks improve safety, and whether paid tiers create sustainable revenue.

The analytics platform is therefore split into five domains: product analytics, weather verification analytics, safety analytics, commercial growth analytics, and operational intelligence. These domains share a common event pipeline, warehouse model, privacy layer and dashboarding framework, but each has different data retention, data quality and governance requirements.

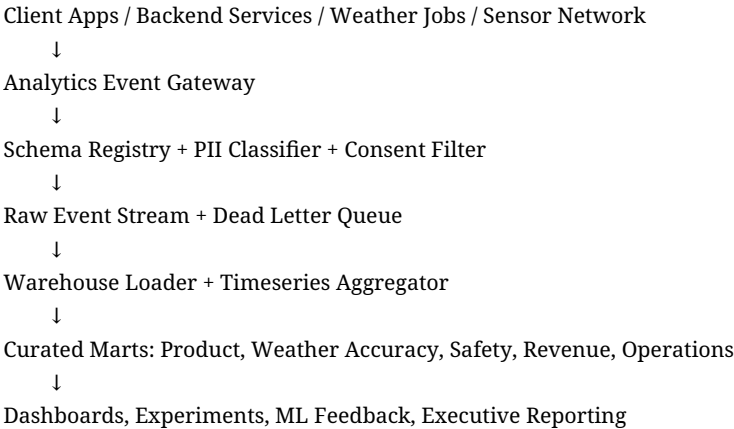
- Product analytics answers: are pilots finding sites, reading briefings, setting alerts and returning to the app?
- Forecast analytics answers: did the model predict the actual wind, gust, cloudbase and thermal behavior?
- Safety analytics answers: did the app avoid false-green recommendations and warn pilots early enough?
- Commercial analytics answers: which features drive conversion, retention, renewals and upgrades?
- Operational analytics answers: are ingestion, APIs, WebSockets, notifications, billing and mobile sync reliable?

2. Analytics Architecture Principles

Principle	Implementation Rule	Reason
Safety before growth	Never optimize conversion by hiding uncertainty, risk or NO-GO messages	The product supports aviation-adjacent safety decisions
Measure decisions, not vanity clicks	Primary metrics must include drive decision, launch decision, alert action and safety outcomes	Paragliding users judge value by real-world decisions
Privacy by design	Precise GPS tracks are pseudonymized, retention-controlled and opt-in for analytics	Location data is sensitive personal data
Ground truth separation	Forecast outputs, real observations and user reports are stored separately	Prevents circular validation and ML leakage
Traceable recommendations	Every AI/flyability output must link to source forecast, launch rule and confidence input	Required for audits and user trust
Tier-aware analytics	Feature usage is tagged with entitlement and plan context	Enables pricing decisions and entitlement enforcement
Experiment safety guardrails	No experiment can weaken hard safety blockers or legal disclaimers	Prevents growth tests from creating safety risk

3. Logical Analytics System Overview

The analytics system is implemented as an event-driven data platform. Client applications and backend services emit structured events. Events pass through a privacy and validation gateway before being written to raw event storage and transformed into warehouse models. Dashboards, ML feature feedback, growth analysis and safety reporting consume the curated data marts.



3.1 Analytics Platform Components

Component	Responsibilities	Primary Owner
analytics-event-gateway	Accept events, validate schema, attach tenant/user/session metadata, enforce consent and rate limits	Data platform
schema-registry	Version event schemas, validate breaking changes and manage client SDK compatibility	Data platform
consent-filter	Drop or pseudonymize analytics fields according to privacy settings and region	Security/privacy
event-stream	Durable event log for product and system events	Platform engineering
warehouse-loader	Batch/micro-batch loader into analytics warehouse	Data engineering
metrics-engine	Computes standardized KPIs, retention cohorts, accuracy metrics and revenue metrics	Analytics
experiment-service	Assigns users/orgs/sites to experiments while enforcing safety guardrails	Product growth
dashboard-layer	Provides curated dashboards for product, safety, clubs, schools, events and ops	Analytics
data-quality-monitor	Detects missing events, abnormal volumes, schema drift and late-arriving data	Data engineering

4. Event Taxonomy

4.1 Event Naming Standard

Events use a stable noun_verb convention. The first token identifies the domain, the second token identifies the object, and the final token identifies the action. Names are lowercase snake_case and must not contain UI-specific naming unless the event is explicitly a UI event.

- Good:
- site_forecast_viewed
 - flyability_decision_opened
 - alert_rule_created

in_flight_airspace_warning_triggered
subscription_checkout_started
forecast_error_sample_created

Bad:
clicked_blue_button
user_did_thing
opened_page_v3_experiment

4.2 Required Event Envelope

```
{
  "event_id": "uuid",
  "event_name": "site_forecast_viewed",
  "schema_version": "1.0.0",
  "occurred_at": "2026-06-05T11:22:33Z",
  "received_at": "2026-06-05T11:22:34Z",
  "source": "mobile|web|backend|worker|sensor|api",
  "environment": "dev|staging|prod",
  "anonymous_id": "uuid",
  "user_id_hash": "optional sha256",
  "org_id": "optional uuid",
  "session_id": "uuid",
  "device_id_hash": "optional sha256",
  "plan_tier": "free|pro|pro_plus|club|school|event|rescue|api",
  "consent_scope": ["essential", "product_analytics"],
  "geo_precision": "none|country|region|coarse|exact",
  "properties": {}
}
```

4.3 Event Categories

Category	Examples	Retention Default	PII Risk
Product navigation	app_opened, map_layer_toggled, site_search_performed	24 months	Low
Decision intelligence	flyability_decision_opened, drive_decision_viewed	36 months	Medium
Weather verification	forecast_error_sample_created, station_observation_ingested	60 months	Low
Safety	alert_triggered, no_go_shown, sos_started, airspace_warning_triggered	60 months or legal policy	High
Commercial	checkout_started, subscription_created, entitlement_blocked	36 months	Medium
Community	pilot_observation_submitted, playbook_change_requested	36 months	Medium
Operations	job_failed, api_latency_sampled, notification_delivered	12 months	Low
Experimentation	experiment_assigned, variant_exposed, metric Converted	24 months	Medium

5. Product Analytics Event Plan

Product analytics measures whether users move from weather exploration to confident decisions. The platform must be able to reconstruct a user journey from initial site discovery through forecast review, AI briefing, alert setup, drive decision and post-flight report.

Event	Trigger	Key Properties	Primary Metric
-------	---------	----------------	----------------

app_opened	Mobile or web app session starts	entry_surface, app_version, locale	DAU/MAU
site_search_performed	User searches for launch/site/location	query_type, result_count, search_radius_km	Site discovery success
site_forecast_viewed	User opens a launch forecast	site_id, source_surface, forecast_hour	Forecast engagement
map_layer_toggled	User changes weather layer	layer_id, enabled, altitude_m, entitlement_required	Layer usage
timeline_scrubbed	User moves time slider	from_time, to_time, layer_id	Forecast exploration depth
flyability_decision_opened	User views GO/MAYBE/NO-GO reasoning	site_id, decision, confidence	Decision engagement
ai_briefing_requested	User asks AI flight briefing	site_id, context_size, plan_tier	AI value
alert_rule_created	User creates flyability/weather alert	rule_type, site_id, threshold	Alert activation
drive_decision_viewed	User opens drive recommendation	drive_time_min, decision, confidence	Trip-decision adoption
pilot_observation_submitted	User reports reality	site_id, wind_kmh, cloudbase_m, thermals_ms	Reality layer contribution

6. Weather Accuracy Analytics

Weather accuracy analytics is one of the main competitive advantages. It measures forecast skill at the places pilots actually care about: launch, landing, ridge, valley and route segments. Accuracy must be computed by variable, model, altitude, terrain class, time horizon, season and site.

6.1 Forecast Error Sample Model

```
forecast_error_sample = {
  "sample_id": "uuid",
  "site_id": "uuid",
  "model_run_id": "uuid",
  "model_name": "ecmwf|gfs|metno|open_meteo|corrected",
  "forecast_valid_time": "timestamp",
  "lead_time_hours": 12,
  "variable": "wind_speed_kmh|gust_kmh|wind_dir_deg|cloudbase_m|thermal_ms",
  "altitude_m_agl": 1000,
  "forecast_value": 18.4,
  "observed_value": 22.1,
  "error": -3.7,
  "abs_error": 3.7,
  "observation_source": "station|pilot_report|track_inference|webcam_cv",
  "quality_score": 0.92
}
```

6.2 Forecast Metrics

Metric	Formula / Meaning	Used For
MAE	Average absolute error by variable/site/model	Primary model comparison
RMSE	Penalizes large forecast errors	Detecting dangerous model failures
Bias	Mean forecast - observed value	Detecting systematic over/under prediction
Wind direction circular error	Circular angular difference in degrees	Launch compatibility validation
False green rate	App says GO but observed hard blocker occurred	Safety-critical release gate
False red rate	App says NO-GO but actual observed conditions were safe	Usability and trust metric
Model disagreement index	Spread between models normalized by variable	Confidence engine
Observation coverage	% of site-hours with ground truth	Digital twin maturity
Correction lift	Error reduction of corrected forecast vs base model	ML value proof

6.3 Site Accuracy Dashboard

- Filter by site, region, season, model, altitude, lead time and variable.
- Show base model vs corrected model MAE for wind, gust, cloudbase and thermal potential.
- Show forecast confidence calibration: when confidence is 80%, conditions should validate roughly 80% of the time.
- Show launch-specific failure modes: strong underforecast in W wind, high valley-wind bias after 15:00, cloudbase overprediction in spring.
- Feed digital twin backlog with sites where correction lift is high or false-green rate is unacceptable.

7. Safety Analytics

Safety analytics is the governance layer for all recommendation systems. It must detect when the product is producing unsafe optimism, delayed warnings, confusing wording or poor uncertainty communication.

7.1 Safety KPI Matrix

KPI	Definition	Target / Gate	Owner
False green rate	GO recommendation followed by observed hard blocker within decision window	Must trend toward zero; release blocker if above threshold	Safety committee
Late warning rate	Hazard appears but alert arrives too late for reasonable action	Release blocker for in-flight safety assistant	Safety + mobile
NO DATA correctness	System shows NO DATA when required inputs missing	100% for hard blockers	Backend
AI unsupported claim rate	AI briefing includes weather/safety claim not grounded in source context	0 allowed in production evals	AI/ML
User report conflict rate	Pilot reports contradict current app recommendation	Investigation queue	Community ops
Airspace alert precision	Warnings triggered for actual relevant route/altitude conflicts	High precision required to reduce alert fatigue	Tracking
SOS delivery success	Emergency share link/notification delivered to contact/rescue channel	99%+ for supported channels	Platform ops

7.2 Safety Event Examples

in_flight_weather_warning_triggered
properties:
 flight_id
 hazard_type: gust_increase | storm_cell | cloudbase_drop | rotor_zone | airspace
 severity: advisory | warning | critical
 lead_time_min
 source: nowcast | station | radar | lightning | forecast_update
 user_action: acknowledged | ignored | landed | unknown

flyability_decision_audited
properties:
 site_id
 decision: GO | MAYBE | NO_GO | NO_DATA
 confidence
 hard_blockers_present
 observed_outcome: safe | marginal | unsafe | unknown
 false_green: boolean

8. Growth and Subscription Analytics

Growth analytics must connect commercial conversion to actual pilot value. The subscription model should not simply count paywall impressions. It must identify which premium capabilities drive real decisions: model comparison, AI briefing, forecast correction, alert rules, offline maps, in-flight safety, school portals and club operations.

8.1 SaaS Metrics

Metric	Definition	Calculation Notes
Visitor to signup	Anonymous visitors who create account	Segment by source and country
Signup to activated pilot	New account that views 3 site forecasts or creates 1 alert within 7 days	Primary activation metric
Free to paid conversion	Free users starting subscription	Segment by trigger feature
MRR / ARR	Monthly and annual recurring revenue	Separate individual vs organization accounts
ARPU / ARPA	Average revenue per user/account	Club/school/event treated as accounts
Expansion MRR	Upgrades from Pro to Pro+ or org seat expansion	Key B2B growth metric
Churn	Cancellation or non-renewal	Use cohort and revenue-weighted churn
LTV:CAC	Lifetime value divided by acquisition cost	Investor metric
Paywall conversion by feature	% users converting after entitlement block	Do not count safety-blocked features as growth wins

8.2 Funnel Definitions

Individual Pilot Funnel:

- anonymous_visit
- > account_created
- > first_site_forecast_viewed
- > first_flyability_decision_opened
- > first_alert_rule_created
- > premium_feature_attempted
- > checkout_started
- > subscription_created
- > retained_30d

Club/School Funnel:

- org_demo_requested
- > org_workspace_created
- > launch_site_claimed
- > member_invited
- > playbook_submitted
- > portal_active_14d
- > subscription_created
- > renewal_successful

9. Analytics Data Model

9.1 Warehouse Star Schema

The analytics warehouse uses a hybrid star schema. Raw immutable events remain available for replay, while curated fact and dimension tables support dashboards and machine learning feedback loops. TimescaleDB handles high-frequency observations; the warehouse handles aggregated analytics.

Table	Grain	Purpose
fact_product_events	One validated analytics event	General product behavior
fact_site_forecast_views	One forecast view per user/session/site/time	Site engagement and conversion
fact_flyability_decisions	One decision view/output	Safety, trust and product value
fact_alert_deliveries	One notification attempt	Alert reliability and effectiveness
fact_forecast_errors	One forecast vs observation comparison	Model accuracy and correction lift
fact_subscription_events	One billing/subscription event	Revenue analytics

fact_experiment_exposures	One exposure to experiment variant	A/B testing
dim_user	Pseudonymized user/account attributes	Cohorts and segmentation
dim_site	Launch site attributes and region	Site analytics
dim_plan	Subscription tier and entitlement bundle	Revenue segmentation
dim_model_run	Forecast model run metadata	Weather verification

9.2 SQL Seed Examples

```
CREATE TABLE analytics_events_raw (
  event_id UUID PRIMARY KEY,
  event_name TEXT NOT NULL,
  schema_version TEXT NOT NULL,
  occurred_at TIMESTAMPTZ NOT NULL,
  received_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  source TEXT NOT NULL,
  environment TEXT NOT NULL,
  anonymous_id UUID,
  user_id_hash TEXT,
  org_id UUID,
  session_id UUID,
  plan_tier TEXT,
  consent_scope TEXT[] NOT NULL DEFAULT '{}',
  geo_precision TEXT NOT NULL DEFAULT 'none',
  properties JSONB NOT NULL DEFAULT '{}',
  validation_status TEXT NOT NULL DEFAULT 'accepted'
);
```

```
CREATE INDEX idx_analytics_events_name_time
  ON analytics_events_raw (event_name, occurred_at DESC);
CREATE INDEX idx_analytics_events_props_gin
  ON analytics_events_raw USING GIN (properties);
```

```
CREATE TABLE fact_forecast_errors (
  sample_id UUID PRIMARY KEY,
  site_id UUID NOT NULL,
  model_run_id UUID,
  model_name TEXT NOT NULL,
  forecast_valid_time TIMESTAMPTZ NOT NULL,
  lead_time_hours INT NOT NULL,
  variable TEXT NOT NULL,
  altitude_m_agl INT,
  forecast_value DOUBLE PRECISION,
  observed_value DOUBLE PRECISION,
  error DOUBLE PRECISION,
  abs_error DOUBLE PRECISION,
  observation_source TEXT,
  quality_score DOUBLE PRECISION,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

```
CREATE INDEX idx_forecast_errors_site_variable_time
  ON fact_forecast_errors (site_id, variable, forecast_valid_time DESC);
```

10. Event Gateway API

The event gateway is the only public event collection surface. It must reject unknown schemas, apply consent rules, enforce rate limits and record validation failures. Internal services may emit to the same gateway or directly to the stream through authenticated service credentials.

POST /v1/analytics/events
Authorization: Bearer <access_token optional for anonymous>
Idempotency-Key: <event_id>

Request:

```
{
  "events": [
    {
      "event_id": "2ef1...",
      "event_name": "map_layer_toggled",
      "schema_version": "1.0.0",
      "occurred_at": "2026-06-05T12:00:00Z",
      "anonymous_id": "...",
      "session_id": "...",
      "properties": {
        "layer_id": "wind_altitude",
        "altitude_m": 1000,
        "enabled": true,
        "site_id": "optional"
      }
    }
  ]
}
```

Response:

```
{
  "accepted": 1,
  "rejected": 0,
  "rejections": []
}
```

10.1 Event Gateway Validation Rules

Rule	Behavior
Unknown event name	Reject with schema_unknown
Unsupported schema version	Reject unless compatibility adapter exists
Missing consent	Drop non-essential analytics fields or reject non-essential event
Precise location without permission	Coarsen to configured precision or reject
Duplicate idempotency key	Return accepted duplicate without double writing
High event rate	Rate-limit per device/user/IP and send telemetry to abuse monitor
Oversized property payload	Reject and log schema violation

11. Dashboard Architecture

11.1 Executive Dashboard

- North-star metric: weekly pilots who made at least one grounded flight decision.
- MRR/ARR, conversion, churn, paid plan mix and expansion revenue.
- Forecast correction lift vs base model at top 50 sites.
- False-green rate, safety alerts delivered, and unresolved safety incidents.
- DAU/WAU/MAU, 30-day retention and active site coverage.

11.2 Product Dashboard

- Site discovery funnel: search -> result -> forecast -> decision -> alert.
- Layer usage by skill level, region and plan tier.
- AI briefing request rate, answer helpfulness and unsupported-claim audit rate.
- Offline map pack downloads and in-flight mode starts.
- Feature adoption matrix by Free, Pro, Pro+, Club, School and Event plans.

11.3 Weather Accuracy Dashboard

- MAE/RMSE/bias by model, variable, altitude, lead time and site.
- Corrected forecast vs base model skill lift.
- Model disagreement map for tomorrow and weekend flying windows.
- Ground-truth coverage heatmap: stations, pilot reports, track-derived observations and webcam CV.
- Digital twin maturity score per launch.

11.4 Safety Dashboard

- NO-GO blockers triggered by region and season.
- False-green investigations and unresolved audit cases.
- In-flight warnings by severity, type and action taken.
- Airspace warning precision and alert fatigue indicators.
- Emergency/SOS delivery success and rescue-link usage.

12. Experimentation Framework

Experiments are allowed for onboarding, layout, feature education, pricing copy, trial length and notification wording. Experiments are not allowed to weaken hard safety blockers, suppress uncertainty, hide legal disclaimers, downgrade NO-GO conditions, or generate unsupported AI advice.

Experiment Type	Allowed?	Examples	Safety Review Required
Onboarding copy	Yes	Different explanations of flyability score	No unless safety wording changed
Paywall presentation	Yes	Compare feature preview vs direct paywall	No
Map layer defaults	Yes with limits	Default wind altitude to 500m or surface	Yes if risk layers affected
Alert wording	Yes	Short vs detailed wind increase alert	Yes
NO-GO threshold	No for growth experiments	Cannot change hard safety blockers to improve engagement	Not allowed
AI briefing guardrails	No for unsafe variants	Cannot allow non-grounded claims	Not allowed
Pricing	Yes	Monthly vs annual emphasis	No

12.1 Experiment Assignment Contract

GET /v1/experiments/assignments?surface=mobile_home

Response:

```
{
  "assignments": [
    {
      "experiment_key": "onboarding_flyability_explainer_v1",
      "variant": "compact",
      "assignment_id": "uuid",
      "safety_review_id": null,
      "expires_at": "2026-09-01T00:00:00Z"
    }
  ]
}
```

13. Privacy-Aware Analytics

Analytics must support useful product and safety insights without unnecessary location surveillance. The system uses explicit consent scopes, pseudonymized identifiers, configurable geolocation precision, retention schedules and a strict separation between operational safety data and growth analytics.

Data Type	Default Analytics Handling	Retention
Exact GPS track	Not used for growth analytics; stored only if tracking/rescue feature enabled	User configurable; default limited
Coarse region	Allowed for product analytics if consented	24 months
Site forecast view	Allowed with pseudonymous user/session id	36 months
Safety incident / SOS	Restricted operational dataset; not used for marketing segmentation	Policy/legal retention
Payment event	Stored through billing provider references; no card data in analytics	36 months
Pilot report	Stored with user visibility controls and moderation state	36 months
AI prompt text	Minimized; avoid storing raw free text unless needed for safety QA and consented	Short retention

14. Data Quality and Observability

14.1 Data Quality Checks

Check	Detection	Action
Event volume anomaly	Hourly count outside expected band by source/event	Alert data team; annotate dashboards
Schema violation spike	Rejected events > threshold	Block bad client release if severe
Missing forecast validation	No forecast_error_samples for covered site windows	Investigate stations/loader
Revenue mismatch	Billing provider MRR differs from warehouse MRR	Reconcile subscription fact table
Consent drift	Events with disallowed fields after opt-out	Security incident review
Late events	occurred_at significantly before received_at	Route to late-event partition
Duplicate events	Same event_id or idempotency key repeated	Deduplicate and report client bug

14.2 Analytics SLOs

SLO	Target
Event ingestion availability	99.9% monthly for accepted clients
Dashboard freshness - operational	< 5 minutes for ops/safety dashboards
Dashboard freshness - product/revenue	< 4 hours for business dashboards
Forecast accuracy mart freshness	Within 1 hour after validation observations arrive
Event schema validation latency	p95 < 100 ms per batch
Warehouse load success	99% daily jobs successful or automatically retried

15. Analytics for Digital Twin and ML Feedback

Analytics must close the loop between observed reality and forecast improvement. Product usage, pilot observations and sensor measurements become training and evaluation signals only after quality filtering, consent checks and leakage prevention. ML training data must not include features that would not be known at prediction time.

Feedback Loop:

- forecast_output_created
- > pilot views decision
- > station/pilot/track observation arrives
- > forecast_error_sample_created
- > model skill dashboard updates
- > correction model training dataset refreshed
- > candidate model evaluated

-> safety gate checks false-green rate

-> model promoted or rejected

- Never train forecast correction on future observations as input features for live predictions.
- Track feature availability time and observation time separately.
- Use blocked time-series validation by site and season.
- Report correction lift separately for wind, gust, cloudbase, thermals and risk outcomes.
- Model promotion requires better accuracy and no degradation in safety metrics.

16. Growth Intelligence Use Cases

16.1 Drive Decision Optimization

The Drive or Don't Drive feature is a key commercial differentiator. Analytics should measure how often it prevents wasted trips, how often it leads to successful flying days and whether users trust its confidence score.

Signal	Event Source	Business Value
Drive decision viewed	drive_decision_viewed	Feature adoption
Navigation opened after GO	external_navigation_opened	High intent
User later submits flight/report	pilot_observation_submitted or track_uploaded	Decision validation
User cancels after MAYBE/NO-GO	no direct telemetry; infer if no trip action	Potential safety/value metric
Drive confidence vs observed outcome	forecast_error_sample + user action	Trust calibration

16.2 Paid Feature Value Tracking

Premium Feature	Value Metric	Conversion Signal
Model comparison	User opens disagreement panel before deciding	Pro+ paywall conversion
AI briefing	Repeat AI requests and helpfulness rating	Pro conversion
Advanced alerts	Alert created and later triggered usefully	Retention
Offline maps	Map pack downloaded before flight mode	Mobile Pro value
In-flight safety assistant	Flight sessions with acknowledged warning	Pro+ safety value
Club playbooks	Sites with approved playbook and member views	Club renewal
School session planner	Training session created and weather-approved	School conversion
Weather station dashboard	Station owners checking health/live wind	Hardware subscription

17. Analytics APIs and Data Access

17.1 Internal Metrics API

GET /v1/internal/metrics/kpi-summary?from=2026-06-01&to=2026-06-30

Authorization: Bearer <internal_token>

Response:

```
{
  "active_decision_users": 18420,
  "free_to_paid_conversion": 0.047,
  "false_green_rate": 0.0012,
  "forecast_correction_lift": {
    "wind_speed_kmh": 0.18,
    "gust_kmh": 0.11,
    "cloudbase_m": 0.09
  },
  "mrr_eur": 48200,
```

```
"alert_delivery_success": 0.995
}
```

17.2 Club/School Analytics API

```
GET /v1/orgs/{org_id}/analytics/overview
Response:
{
  "org_id": "uuid",
  "period": "last_30_days",
  "member_active_count": 42,
  "site_forecast_views": 318,
  "approved_playbook_views": 146,
  "alerts_triggered": 27,
  "training_sessions_planned": 12,
  "no_go_sessions_blocked": 3,
  "weather_station_uptime": 0.982
}
```

18. Roles and Access Control

Role	Analytics Access
Individual pilot	Own usage history, alert effectiveness, personal flight/weather summaries if enabled
Club admin	Aggregated organization analytics, no precise member tracks without explicit sharing
School admin/instructor	Training-session weather analytics and student progress visibility according to role
Event organizer	Event task analytics, weather windows, pilot tracking only for event consent scope
Rescue operator	Emergency/rescue dashboard only during authorized incident workflow
Product manager	Aggregated product analytics; no raw sensitive location unless specifically approved
Data scientist	Pseudonymized datasets; access logged and scoped
Security/privacy officer	Consent, deletion, audit and incident analytics
Executive/investor reporting	Aggregated KPIs only

19. Implementation Plan

19.1 MVP Analytics Slice

1. Implement analytics event gateway with schema validation and consent filtering.
2. Track app_opened, site_search_performed, site_forecast_viewed, map_layer_toggled, flyability_decision_opened, alert_rule_created and subscription_checkout_started.
3. Create raw event table and curated daily KPI views.
4. Implement forecast_error_sample generation for wind/gust/cloudbase at MVP launch sites.
5. Create executive, product, weather accuracy and safety dashboards.
6. Add basic paywall conversion and subscription metrics.
7. Create data quality alerts for missing events and schema violations.

19.2 V1/V2 Roadmap

Phase	Analytics Capabilities
MVP	Core events, product funnels, basic forecast accuracy, basic revenue metrics
V1	Cohorts, retention, alert effectiveness, model disagreement dashboards, paid feature value tracking
V2	Experiment platform, digital twin maturity scoring, AI safety eval

	dashboard, org analytics
V3	Predictive churn, lifetime value modeling, automated growth insights, station ROI analytics
V4	Global benchmark network, anonymous regional safety analytics and automated launch-quality scoring

20. Engineering Backlog

ID	Epic	Task	Priority
AN-001	Gateway	Build /v1/analytics/events endpoint with schema validation	P0
AN-002	Schema	Create analytics schema registry and compatibility checks	P0
AN-003	Privacy	Implement consent filter and geo precision controls	P0
AN-004	Warehouse	Create raw event and curated fact tables	P0
AN-005	Weather Accuracy	Generate forecast_error_sample pipeline	P0
AN-006	Dashboards	Build executive/product/weather/safety dashboards	P1
AN-007	Revenue	Integrate billing/subscription events into warehouse	P1
AN-008	Experimentation	Build experiment assignment service with safety guardrails	P2
AN-009	Org Analytics	Build club/school/event analytics dashboards	P2
AN-010	ML Feedback	Connect forecast error marts to ML feature store and model registry	P2
AN-011	Quality	Automated data quality tests and event anomaly alerts	P1
AN-012	Exports	Build safe aggregated investor/reporting exports	P3

21. Acceptance Criteria

- All production events conform to a registered schema version.
- A user can opt out of non-essential analytics and event gateway respects the consent state.
- Dashboards show DAU, WAU, activation, retention, conversion, churn, MRR and paid feature adoption.
- Weather accuracy dashboard computes MAE, RMSE, bias and false-green rate by site/model/variable.
- Safety dashboard flags false-green cases and unsupported AI weather claims.
- Revenue metrics reconcile with billing provider totals within defined tolerance.
- Data quality alerts fire when event volume, schema errors, revenue mismatch or missing forecast validation exceeds threshold.
- No growth experiment can bypass hard safety blockers or change NO-GO criteria without formal safety approval.

22. Conclusion

Volume 18 turns SkyThermal from a weather visualization product into a measurable decision-intelligence business. The analytics system proves whether pilots trust the product, whether forecasts are improving, whether safety rules are working, whether paid tiers create durable revenue, and where the platform should invest next. This volume should be implemented in parallel with the API, mobile, weather and ML volumes because instrumentation must be designed into the product from the beginning, not added after launch.

Appendix A - Core KPI Dictionary

KPI	Definition	Primary Source
Active decision user	User who opened at least one grounded flyability/drive/AI decision in period	fact_flyability_decisions
Activation	New user who views >=3 site forecasts or creates 1 alert within 7 days	fact_product_events
Decision confidence calibration	Relationship between confidence score and observed safe/marginal/unsafe outcome	fact_forecast_errors + fact_flyability_decisions
Digital twin maturity	Coverage score based on observations, forecast errors, reports, tracks and playbook completeness	site analytics mart
Safety false-green rate	GO recommendation followed by observed hard blocker	safety mart
Correction lift	Reduction in error for corrected forecast vs base model	weather accuracy mart
Paid value realization	Premium feature used in a decision workflow within 30 days of subscription	product + billing mart
Alert effectiveness	Alert delivered, opened/acknowledged and followed by relevant action or safe outcome	alert mart